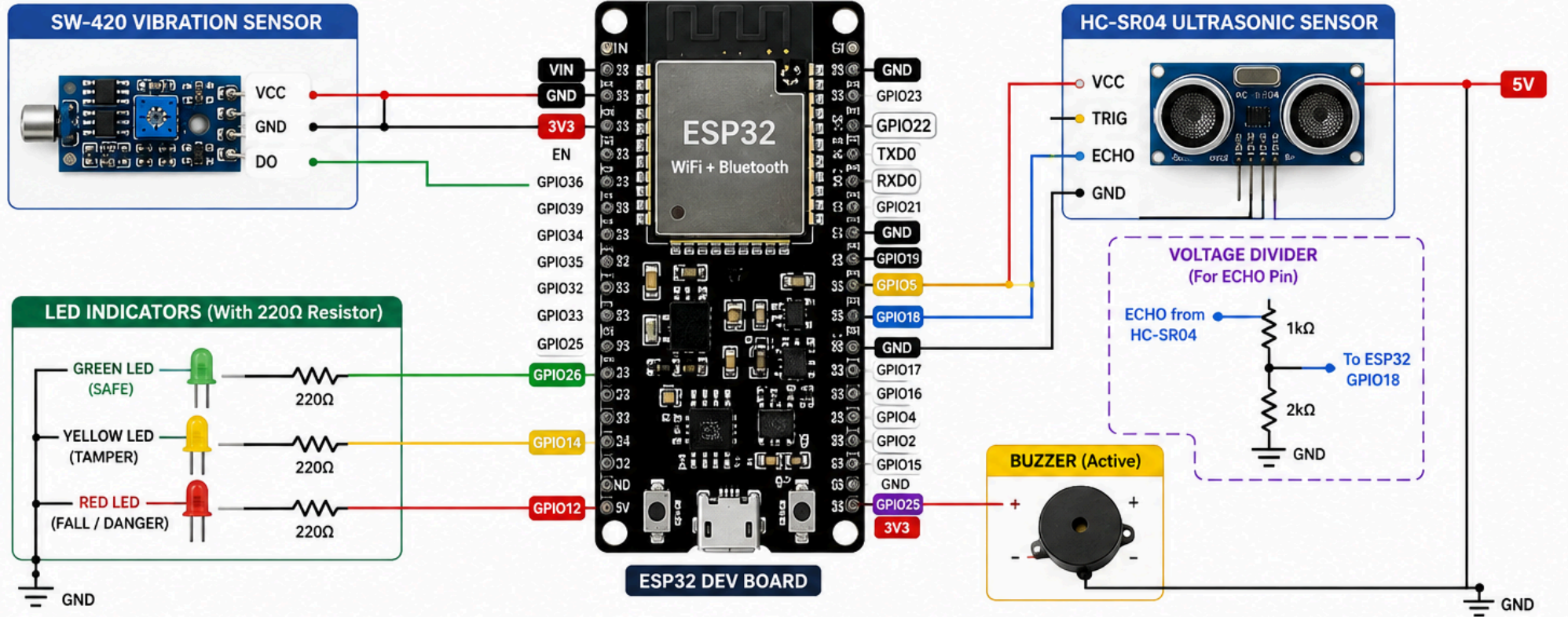


WELSENTRY – CONNECTION DIAGRAM

ESP32 + SW-420 (Vibration) + HC-SR04 (Ultrasonic) + Buzzer + LEDs



CONNECTION SUMMARY			
Component	Signal	ESP32 Pin	Voltage
SW-420	DO	GPIO27	3.3V
HC-SR04	TRIG	GPIO5	3.3V
HC-SR04	ECHO (via divider)	GPIO18	3.3V
Buzzer	+	GPIO25	3.3V
Green LED (Safe)	+	GPIO26	3.3V
Yellow LED (Tamper)	+	GPIO14	3.3V
Red LED (Fall/Danger)	+	GPIO12	3.3V
Common Ground	GND	GND	—

NOTES
- All GND pins are common. - HC-SR04 VCC should be 5V. - ECHO pin of HC-SR04 must be level shifted to 3.3V using the voltage divider. - Use 220Ω resistors with all LEDs. - Buzzer used is active type.

LEGEND
— 5V Power
— GND
— Digital Signal (Sensor)
— Control Signal (TRIG)
— Echo Signal (via Divider)
— Buzzer Signal
— 220Ω Resistor

